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Greenhouse gas emissions from agrifood systems

Global, regional and country trends, 2001–2023

HIGHLIGHTS

- Global agrifood systems emissions reached 16.5 billion tonnes of carbon dioxide equivalent (Gt CO₂eq) in 2023, up 21 percent since 2001. Their share in total emissions fell from 38 to 32 percent in 2023.
- Farm-gate emissions from crop and livestock amounted to 8.1 Gt CO₂eq in 2023, or 49 percent of agrifood systems emissions, marking a 17 percent increase since 2001.
- Pre and post agricultural production emissions rose by 33 percent since 2001 to 5.2 Gt CO₂eq in 2023, accounting for 32 percent of agrifood systems emissions. Emissions from manufacturing, transport, packaging, retail, and household consumption grew by about 80 percent.
- Land-use change emissions declined 6 percent to 3.2 Gt CO₂eq since 2021, accounting for 19 percent of agrifood systems emissions.
- In 2023, livestock emissions (4.3 Gt CO₂eq) were the largest single component, followed by deforestation (2.8 Gt CO₂eq), and packaging, transport and retail (1.4 Gt CO₂eq).
- In 2023, emissions were largest in Asia (7.1 Gt CO₂eq), followed by the Americas (4.8 Gt CO₂eq), Africa (2.4 Gt CO₂eq), Europe (1.9 Gt CO₂eq) and Oceania (0.4 Gt CO₂eq). Since 2001, emissions increased in Asia (+53 percent), Africa (+17 percent), and the Americas (+7 percent), but declined in Europe (–6 percent) and Oceania (–19 percent).
- The global emissions intensity of agricultural production in 2023 was 1.9 kg CO₂eq per international dollar, down 25 percent from 2001. It declined in all regions, from –21 percent in Europe to –31 percent in Oceania.
- Per capita agrifood systems emissions fell by 6 percent since 2001 to 2.0 t CO₂eq/cap in 2023. Oceania remained the highest emitter (8.1 t CO₂eq/cap), followed by the Americas (4.6 t CO₂eq/cap), Europe (2.5 t CO₂eq/cap), Africa (1.6 t CO₂eq/cap) and Asia (1.5 t CO₂eq/cap).

FAOSTAT AGRIFOOD SYSTEMS EMISSIONS

BACKGROUND

Agrifood systems account for about one-third of total anthropogenic greenhouse gas (GHG) emissions. They are generated within the *farm gate*, from crop and livestock production activities; by *land use change*, caused by deforestation, biomass fires and peatland degradation processes often linked to land



clearance for agriculture; and in *pre and post agricultural production processes*, comprising the supply chain including food manufacturing, retail, household consumption and food disposal (Tubiello *et al.*, 2022).

Statistics on the underlying activity data, emissions and indicators (shares of total economy, per capita emissions and emissions intensities) are disseminated in FAOSTAT at the country, regional and global levels, covering over 200 countries and territories, for the period 1961–2023 (FAO, 2025a; FAO, 2025b), together with a comprehensive summary of emissions from the rest of the economy. Data are provided by single component gases – carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases (F-gases) – as well as in carbon dioxide equivalents (CO₂eq). This analysis focuses on the period 2001–2023.

GLOBAL

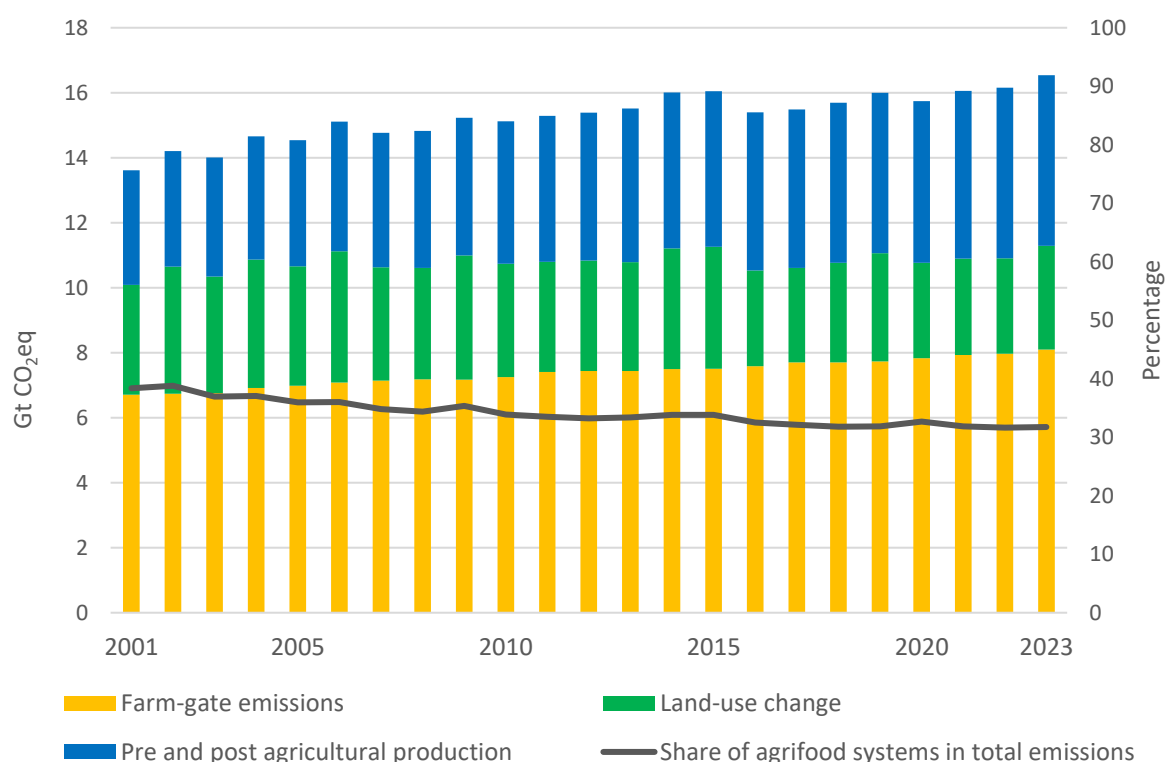
In 2023, agrifood systems emissions were 16.5 billion tonnes of CO₂ equivalent (Gt CO₂eq) which is 21 percent higher than in 2001, though with little change over the last decade (Figure 1). By comparison, global anthropogenic emissions reached 52.1 Gt CO₂eq in 2023, the highest level on record and a 47 percent increase since 2001. The share of agrifood systems emissions in global anthropogenic emissions has decreased from 38 percent in 2001 to 32 percent in 2023.

Within agrifood systems in 2023, emissions within the farm gate accounted for 8.1 Gt CO₂eq, pre and post agricultural production for 5.2 Gt CO₂eq, and land-use change for 3.2 Gt CO₂eq. Although these levels were fairly similar to those recorded in 2022, farm-gate and pre and post agricultural production emissions increased by 17 and 33 percent, respectively, while land-use change emissions decreased by 6 percent compared with 2001. Farm-gate emissions remained the largest component of agrifood systems in 2023 (49 percent, the same as in 2001), whereas the contribution of land-use change emissions dropped to 19 percent (compared with 25 percent in 2001). Emissions from pre and post agricultural production increased their contribution to total agrifood systems emissions to 32 percent, up from 26 percent in 2001.

In 2023, per capita agrifood systems emissions were 2.0 tonnes of CO₂ equivalent per capita (t CO₂eq/cap), compared with 2.2 t CO₂eq/cap in 2001; emissions per value of agricultural production decreased to 1.9 kilograms of CO₂ equivalent per international dollar (kg CO₂eq/US\$) in 2023 compared with 2.5 kg CO₂eq/US\$ in 2001, reflecting increased production efficiencies in relation to inputs and resource use. Finally, emissions per area of agricultural land were 2.4 tonnes of CO₂ equivalent per hectare (t CO₂eq/ha), compared with 2.1 t CO₂eq/ha in 2001.



Figure 1: Global agrifood systems emissions by component and share of agrifood systems emissions in total emissions



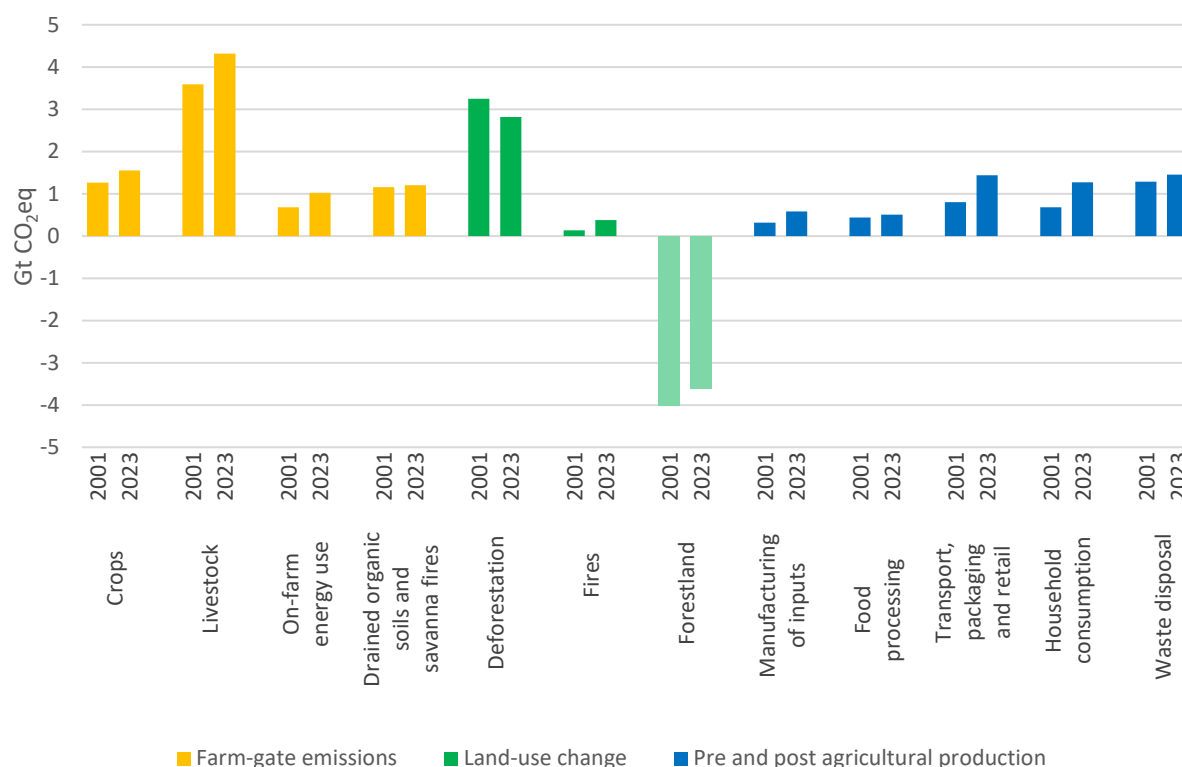
Source: FAO. 2025. FAOSTAT: Emissions totals. [Accessed October 2025].

<https://www.fao.org/faostat/en/#data/GT>. Licence: CC-BY-4.0; and FAO. 2025. FAOSTAT: Emissions indicators. [Accessed November 2025]. <https://www.fao.org/faostat/en/#data/EM>. Licence: CC-BY-4.0.

In 2023, the largest component of agrifood systems emissions globally was livestock, totalling 4.3 Gt CO₂eq from methane generated in enteric fermentation and nitrous oxide from manure. The second most important component was deforestation, at 2.8 Gt CO₂eq from biomass carbon losses, while the third was packaging, transport and retail, generating 1.4 Gt CO₂eq largely from fossil fuel combustion across logistics and distribution systems (Figure 2).

Compared with 2001, livestock emissions grew by 22 percent, while deforestation emissions decreased by 13 percent. This suggests a decoupling over time between the two processes, reflecting known decadal trends towards more intensive livestock systems, needing less land clearance for grazing. At the same time, the strongest growth was recorded in supply chain and consumption processes. Emissions from the manufacturing of inputs, transport, packaging and retail, and household consumption grew by about 80 percent from 2001 to 2023.

Figure 2: Agrifood systems emissions by component



Note: Emissions/removals on forestland are not accounted for within agrifood systems in this analysis and are shown here for comparison. They are disseminated in FAOSTAT as part of land-use, land-use change and forestry (LULUCF) emissions.

Source: FAO. 2025. FAOSTAT: Emissions totals. [Accessed October 2025].

<https://www.fao.org/faostat/en/#data/GT>. Licence: CC-BY-4.0.

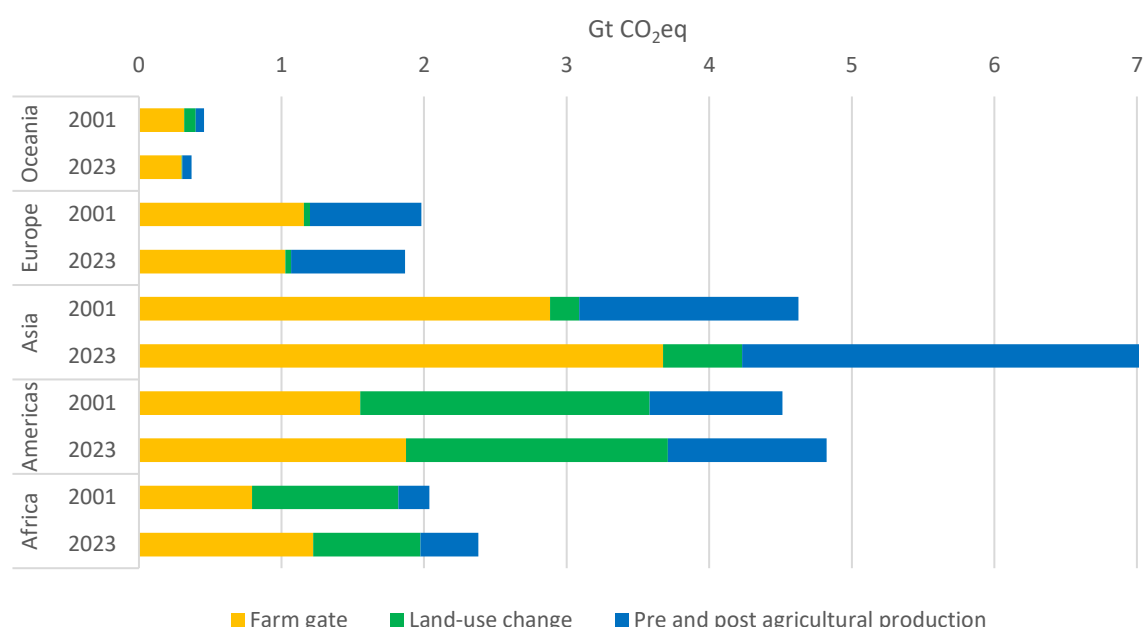
REGIONAL

In 2023, Asia was the largest contributor to regional agrifood systems emissions with 7.1 Gt CO₂eq, followed by the Americas (4.8 Gt CO₂eq), Africa (2.4 Gt CO₂eq), Europe (1.9 Gt CO₂eq), and Oceania (0.4 Gt CO₂eq) (Figure 3), largely reflecting the relative size of the regions. Emissions produced within the farm gate accounted for more than half of agrifood systems emissions across all regions except the Americas, with a particularly high share in Oceania (81 percent) compared with Europe (55 percent), Asia (52 percent), and Africa (51 percent). In contrast, farm-gate emissions accounted for 39 percent of the total agrifood systems emissions in the Americas, slightly more than those from land-use change (38 percent).

Emissions in Africa and the Americas had significant land-use change components of agrifood systems emissions (respectively 0.8 and 1.8 Gt CO₂eq), representing 43 percent and 31 percent, respectively, of their total agrifood systems emissions. In contrast, significant pre and post agricultural production emissions were observed in Asia (2.9 Gt CO₂eq; 40 percent) and Europe (0.8 Gt CO₂eq; 43 percent), reflecting higher energy demand for manufacturing, refrigeration and food distribution in these regions.

Compared with 2001, agrifood systems emissions increased by 53 percent in Asia, followed by Africa (17 percent) and the Americas (7 percent), whereas they decreased in Europe (6 percent) and Oceania (19 percent).

Figure 3: Agrifood systems emissions by component and region



Source: FAO. 2025. FAOSTAT: Emissions totals. [Accessed October 2025].

<https://www.fao.org/faostat/en/#data/GT>. Licence: CC-BY-4.0.

Oceania was the largest per capita emitter in 2023 with 8.1 t CO₂eq/cap, followed by the Americas (4.6 t CO₂eq/cap), respectively four and two times the world average per capita emissions. Next were Europe (2.5 t CO₂eq/cap), Africa (1.6 t CO₂eq/cap) and Asia (1.5 t CO₂eq/cap). Emissions per hectare of agricultural land ranged 0.8–3.3 t CO₂eq/ha, with Oceania the lowest per area emitter, and Asia and the Americas the highest. Finally, emissions per value of agricultural production ranged 1.6–4.0 kg CO₂eq/US\$, with Oceania having the highest intensity value and Asia the lowest (Table 1).

Emissions per value of agricultural production decreased across regions by 20–30 percent from 2001. Per capita emissions also decreased between 2001 and 2023, by 8–13 percent in Europe and the Americas, and by 33–43 percent in Africa and Oceania, while they increased in Asia (+22 percent). Farm-gate emissions per agricultural land area decreased in Europe and Oceania (by 5–6 percent), whereas they remained stable in Africa (+1 percent), and increased in the Americas (+5 percent) and Asia (+39 percent).

Table 1: Emissions indicators by region

	Per value of agricultural production			Per capita			Per area of agricultural land		
	2001	2023	Change	2001	2023	Change	2001	2023	Change
	kg/l\$	kg/l\$	%	t/cap	t/cap	%	t/ha	t/ha	%
Africa	3.8	3.0	-22	2.4	1.6	-33	1.7	1.7	1
Americas	2.4	1.9	-20	5.4	4.6	-13	3.1	3.3	5
Asia	2.3	1.6	-29	1.2	1.5	22	1.8	2.6	39
Europe	2.4	1.9	-21	2.7	2.5	-8	2.5	2.3	-6
Oceania	5.8	4.0	-31	14.3	8.1	-43	0.9	0.8	-5
World	2.5	1.9	-25	2.2	2.0	-6	2.1	2.4	14

Source: FAO. 2025. FAOSTAT: Emissions indicators. [Accessed October 2025].

<https://www.fao.org/faostat/en/#data/EM>. Licence: CC-BY-4.0.

COUNTRY

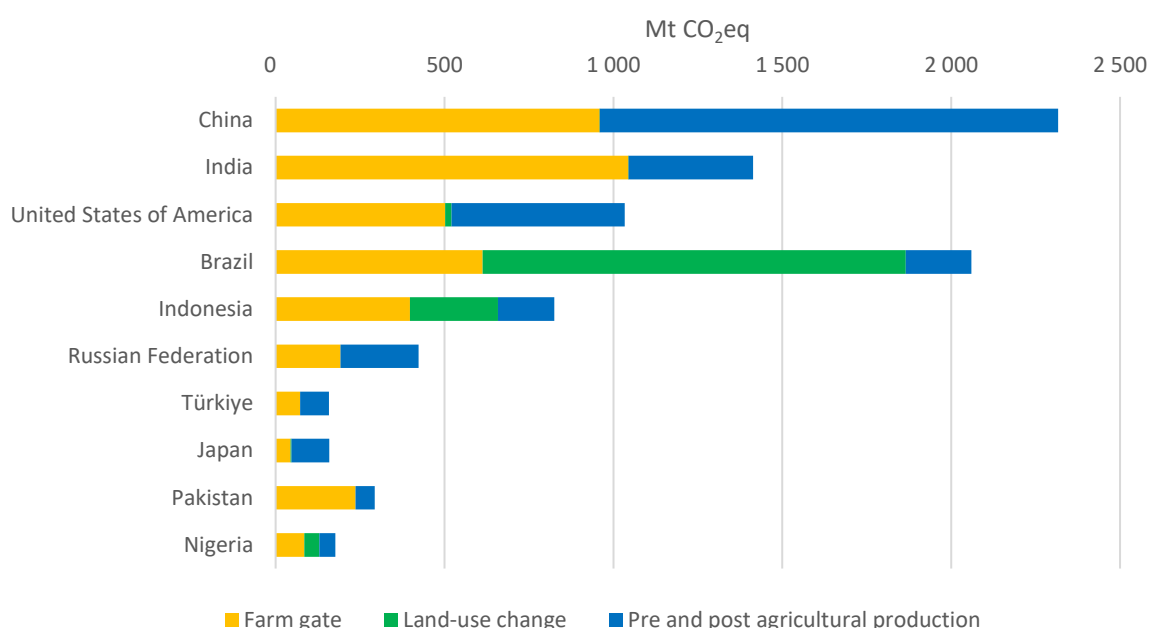
AGRIFOOD SYSTEMS EMISSIONS

Figure 4 presents agrifood systems emissions by component for the countries with the largest value of agricultural production in 2023: China (USD 1 440 billion; 2 317 Mt CO₂eq), India (USD 486 billion; 1 413 Mt CO₂eq), the United States of America (USD 371 billion; 1 034 Mt CO₂eq), Brazil (USD 219 billion; 2 060 Mt CO₂eq), Indonesia (USD 146 billion; 825 Mt CO₂eq), the Russian Federation (USD 98 billion; 423 Mt CO₂eq), Türkiye (USD 94 billion; 158 Mt CO₂eq), Japan (USD 89 billion; 159 Mt CO₂eq), Pakistan (USD 75 billion; 294 Mt CO₂eq) and Nigeria (USD 73 billion; 177 Mt CO₂eq). These countries included seven of the top ten GHG emitters and together accounted for more than half of the 2023 world total agrifood systems emissions.

The agrifood systems emissions composition in the top countries showed three patterns. In Brazil, Indonesia and Nigeria, land-use change played a significant role, contributing 61 percent, 32 percent, and 25 percent of total emissions, respectively. Farm-gate emissions in these countries (respectively 30 percent, 48 percent and 48 percent) exceeded those from pre and post agricultural production activities. Farm-gate emissions were predominant in India and Pakistan (74 percent and 81 percent, respectively), with minimal contributions from land-use change and low shares from pre and post agricultural production. Conversely, pre and post agricultural production emissions were the largest component in China, the United States of America, the Russian Federation, Türkiye and Japan, ranging from 50 percent to 71 percent. This indicates advanced industrialization and extensive energy use throughout their food systems, including cold chains and retail infrastructure. In these countries, land-use change played a minimal role, while farm-gate emissions accounted for a smaller share than pre and post agricultural production emissions.

In 2023, 88 out of 231 countries and territories had supply chain emissions accounting for half or more of the total agrifood systems emissions. High shares were observed in major agricultural producers such as Japan (71 percent), the Republic of Korea (69 percent), and Italy (58 percent). High shares were also observed in many food-importing countries such as Lebanon, Kuwait, Oman, Qatar and Saudi Arabia, each with a share of about 80 percent. The pre and post agricultural production sector accounted for more than 95 percent of national agrifood systems emissions in Several Small Island Developing States such as Aruba, the United States Virgin Islands, Turks and Caicos Islands, and Cayman Islands.

Figure 4: Agrifood systems emissions, top ten countries by value of agricultural production (2023)



Note: Countries are listed in descending order of their value of agricultural production expressed in constant 2014–2016 USD, from China (first) to Nigeria (tenth).

Source: FAO. 2025. FAOSTAT: Emissions Totals. [Accessed October 2025].

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Normalizing country emissions by the population, agricultural area, and value of agricultural production makes them more comparable. Results for 2023 show that agrifood systems emissions indicators varied within much narrower ranges than the corresponding absolute emissions levels. In general, 75 percent of country values (the upper interquartile range) were lower than about twice the median value for each indicator (Table 2).

Table 2: Emissions indicators, summary statistics (2023)

Emissions indicator	Median	Interquartile range	Minimum	Maximum
Per capita (t CO ₂ eq/cap)	1.8	1.2–2.8	0.1	4.2
Per value of agricultural production (kg CO ₂ eq/I\$)	2.0	1.3–3.4	0.5	5.1
Per area of agricultural land (t CO ₂ eq/ha)	2.6	1.5–5.9	0.1	8.8

Note: Minimum and maximum values are computed as 1.5 times the interquartile range, representing in a box plot the lowest and highest data point excluding outliers.

Source: Based on FAO. 2025. FAOSTAT: Emissions indicators. [Accessed October 2025].

<https://www.fao.org/faostat/en/#data/EM>. Licence: CC-BY-4.0.

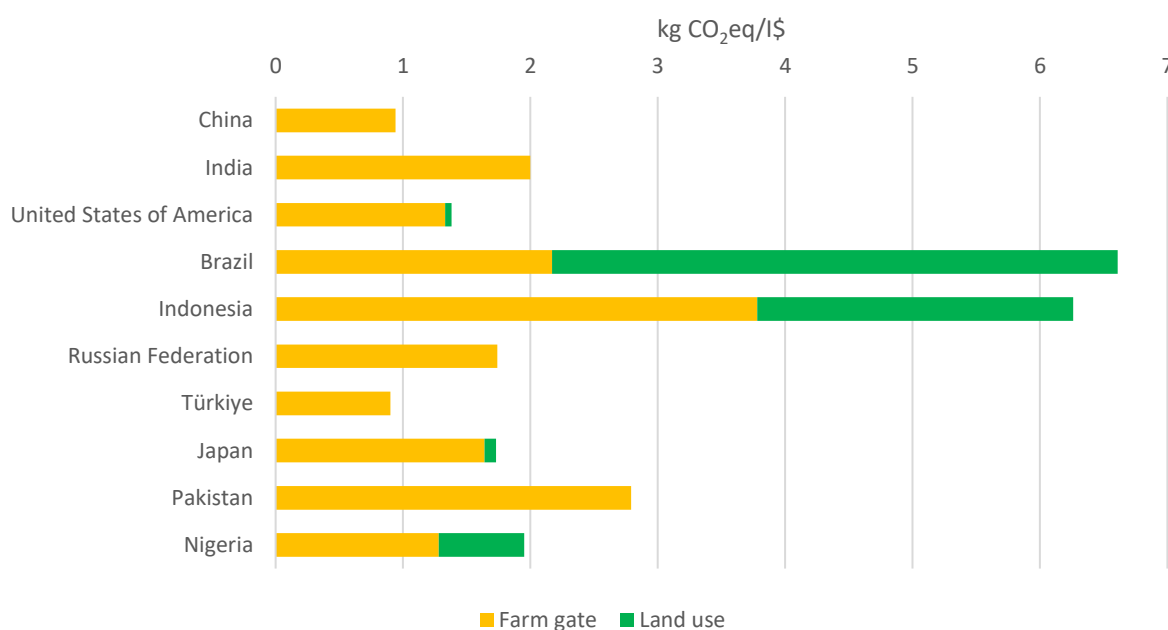
EMISSIONS INTENSITY OF AGRICULTURAL PRODUCTION

In 2023, among the top agricultural producers mentioned in Figure 4, Indonesia (6.3 kg CO₂eq/I\$) and Brazil (6.6 kg CO₂eq/I\$) had the highest emissions intensities (Figure 5), more than double the global median of 2.6 kg CO₂eq/I\$. In contrast, China and Türkiye (about 1.0 kg CO₂eq/I\$ each) were among the world's best performers for this indicator, followed by the United States of America (1.4 kg CO₂eq/I\$). The emissions intensities of Japan, India, Nigeria, the Russian Federation (1.7–2.0 kg CO₂eq/I\$) were below, though close to, the world median, while Pakistan was slightly above it with 2.8 kg CO₂eq/I\$.

The contribution of farm-gate and land-use change processes to the emissions intensity varied across countries. In particular, land-use change significantly contributed to high values in Brazil and Indonesia, reflecting the negative impacts of deforestation and peatland degradation in those countries.

In absolute terms, the countries with high intensities (7–9 kg CO₂eq/I\$) included Peru, the United Republic of Tanzania, Mali, Finland, Timor-Leste, Guinea, Cameroon, Ethiopia, and Paraguay. Conversely, Kiribati, Niue, Ghana, Spain, Sri Lanka, Ukraine, Dominica, Tokelau, Albania, Rwanda, Belgium, the Democratic People's Republic of Korea, and the United States of America had the lowest intensities with values below 1.5 kg CO₂eq/I\$.

Figure 5: Emissions on agricultural land per value of agricultural production, top ten countries by value of agricultural production (2023)



Note: Land use in the figure is calculated as the difference between emissions on agricultural land and farm-gate emissions.

Source: FAO. 2025. FAOSTAT: Emissions indicators. [Accessed October 2025].

<https://www.fao.org/faostat/en/#data/EM>. Licence: CC-BY-4.0.

EXPLANATORY NOTES

The FAOSTAT emissions database is composed of several data domains covering the GHG emissions from agrifood systems. The database includes carbon dioxide (sCO₂) emissions from land-use and land-use change, and from combustion of fossil fuels for pre and post agricultural production processes; non-CO₂ emissions from agricultural activities comprising methane (CH₄) and nitrous oxide (N₂O) emissions; as well as emissions of fluorinated gases (F-gases) used in the agrifood cold chain. The single domains are all summarized in the [Emissions totals](#) domain, where the single-gas emissions are aggregated in CO₂ equivalent (CO₂eq), computed applying the global warming potential from the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC) (IPCC, 2014). In the domain, the single categories of emissions are further summarized by the Food and Agriculture Organization of the United Nations (FAO) aggregates of farm-gate, land-use change and pre and post agricultural production to break down the emissions from agrifood systems, as well as by the categories of the IPCC Agriculture, Forestry and Other Land Use (AFOLU) sector of the national GHG inventories (NGHGI) to the United Nations Framework Convention on Climate Change (UNFCCC).

FAO estimates of emissions from agrifood systems are available by country, regional and global aggregates for the 1961–2023 period for agriculture production processes, i.e. crop and livestock activities. The activity data underlying these emissions are based on country data officially reported to FAO (for instance, livestock numbers, harvested area, fertilizers use in agriculture). Projections to 2030 and 2050 are also available. They are computed with respect to the 2005–2007 baseline, following Alexandratos and Bruinsma (2012).

Land-use and land-use change emissions and removals are instead generally available only for the period 1990–2024. The activity data for forests are collected from FAO [Forest Resources Assessments](#) (FRA) in five-year cycles. Geospatial data complement existing national statistics and provide the source of activity data for emissions on [drained organic soils](#) (1990–2024), [savanna, forest fires and fires in organic soils](#) (1990–2024).

Data on [energy use](#), for all components of pre and post agricultural production are available for the 1990–2023 period. Emissions from pre and post agricultural production processes, including those from [agrifood systems waste disposal](#) are calculated by FAO based on activity data (mostly energy use) from the United Nations Statistics Division (UNSD) and other third parties. For transparency and completeness, [emissions totals](#) integrate information on the emissions from other economic sectors from the PRIMAP-hist dataset v2.4 (Gütschow *et al.*, 2024).

The database disseminates in separate domains the shares of emissions of each category over total emissions. Associated per capita values are also reported in [emissions indicators](#).

For emissions from fires in organic soils, in line with existing literature, only the emissions from South-eastern Asian countries (e.g. Brunei Darussalam, Indonesia and Malaysia) were considered anthropogenic. Conversely, emissions estimates for the other countries and territories provided in FAOSTAT were not considered anthropogenic, to reflect the lack of evidence to this end in existing literature. As a result, although the emissions from fires in organic soils are disseminated for all the countries and territories where these fires occur, the values from countries in the FAOSTAT regional aggregate “South-eastern Asia” only contribute to relevant thematic, regional and world total aggregates.



REFERENCES

- > **Alexandratos, N. & Bruinsma, J.** 2012. *World Agriculture towards 2030/2050: The 2012 Revision*. ESA Working Paper No. 12-03. Rome, FAO. <https://www.fao.org/3/ap106e/ap106e.pdf>
- > **FAO.** 2025a. FAOSTAT: Emission totals. [Accessed October 2025]. <https://www.fao.org/faostat/en/#data/GT>. Licence: CC-BY-4.0.
- > **FAO.** 2025b. FAOSTAT: Emissions shares. [Accessed October 2025]. <https://www.fao.org/faostat/en/#data/EM>. Licence: CC-BY-4.0.
- > **Gütschow, J. & Pflüger, M.** 2024. The PRIMAP-hist national historical emissions time series (1750-2021) v2.4.2. <https://doi.org/10.5281/zenodo.10705513>
- > **Tubiello, F.N., Karl, K., Flammini, A., Gütschow, J., Obli-Laryea, G., Conchedda, G., Pan, X. et al.** 2022. Pre- and Post-Production Processes Increasingly Dominate Greenhouse Gas Emissions from Agri-Food Systems. *Earth System Science Data* 14(4): 1795–1809. <https://doi.org/10.5194/essd-14-1795-2022>

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